

The Shared Circle in Modal Triplet Theory:

Exact Phase Transport, Interpretive Roles, and Physical Boundaries

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Version 2

Abstract

Modal Triplet Theory repeatedly uses a common circle in its bundle, proto-spinor, finite-operator, and compactification descriptions. This paper states what that circle currently means and what it does not yet explain. Its rigorous role is common phase and holonomy data carried by a Hermitian line bundle with connection. On the selected q79 finite symbol, one universal flat line pulls back to the determinant-sign sector, tensors the three trace lanes, and commutes with the finite projectors, complex quarter-turn, and Reynolds-Hessian operator. This is an exact same-source phase-transport result. It neither adds a seventh compact coordinate nor identifies compact phase with Lorentzian time. A physical mass claim still requires a selected Dirac or two-point operator and a dispersion or pole theorem; inertia requires an effective response or worldline coefficient; gravity requires the four-dimensional action and universal stress response; and time requires a Lorentzian evolution and an explicit clock or ordering map. The proposed unification of mass, inertia, gravity, and time is therefore retained as a research program organized by one shared differential line, not as a current derivation of those observables.

Version 2 Revision Note

Supersedes: The first release under the title *The Central Circle: Inertia, Mass, Gravity, and Time as Shared Coherence Bookkeeping in Modal Triplet Theory*.

Reason: A common compact phase was identified directly with mass, inertia, gravity, physical time, family multiplicity, and irreversible dynamics without the required source operators or readout maps.

Resolution: The paper is rebuilt around the exact finite q79 shared-line theorem and gives a separate physical contract for each proposed readout.

Retained result: One flat differential line supplies common connection and holonomy data to the finite q79 lanes and Hessian square without a fitted dimensionless coefficient.

Remaining boundary: The nonflat HYM continuation, physical mass pole, inertial response, four-dimensional stress coupling, clock map, and branch-selection law remain open.

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1 The answer first

The phrase “shared circle” has been used in several ways across the MTT corpus. The current reconciled meaning is:

one source of compact $U(1)$ phase and holonomy data

represented by a Hermitian line bundle and its unitary connection.

This statement is global enough to compare transport around loops. It is stronger than noticing several abstract circles with the same dimension, but weaker than saying that all physical sectors are generated by one coordinate. The circle is counted once. It is not multiplied through the Lens, Nil, and q79 factors, and it is not an extra product dimension.

The exact current theorem belongs to the corrected MTT Foundation. It proves a finite shared-line and Hessian intertwiner on the selected q79 flat root-stack symbol. The present paper explains that result and asks how far it can be transported toward familiar physical concepts.

1.1 Four levels that must not be collapsed

Level	Mathematical object	Current status
Phase fiber	A copy of $U(1)$ in each fiber	Standard construction
Differential line	Hermitian line plus connection and holonomy	Exact on the finite q79 symbol
Continuum coupling	Line coupled to HYM, Dirac, or four-dimensional fields	Partly specified; physical intertwiner open
Observable readout	Mass, inertia, gravity, clock time, or family data	Requires a separate theorem for each readout

The older manuscript moved directly from the first row to the fourth. The finite shared-line result closes the second row and gives a useful source for the third. It does not close the observable rows automatically.

2 Circle fibers, line bundles, and connections

A bare circle S^1 is a one-dimensional compact manifold. In gauge and bundle geometry, the more useful object is a principal $U(1)$ bundle or its associated Hermitian line bundle. Locally it resembles a product $U \times U(1)$, but globally its transition functions can carry nontrivial topology and holonomy.

A unitary connection ∇ tells us how phase is transported. Its curvature records infinitesimal transport around small loops, while holonomy records transport around finite loops. Two line bundles with isomorphic fibers need not have the same topology, connection, or holonomy. Thus the claim that different sectors use “the same circle” is meaningful only after the source line and the connection-preserving pullback maps are named.

2.1 What a shared source guarantees

Suppose a Hermitian line with connection (L, ∇) is defined over a classifying base and sector maps $f_i : X_i \rightarrow B$ are supplied. The sector lines are the pullbacks

$$(L_i, \nabla_i) = f_i^*(L, \nabla).$$

Their curvatures and loop holonomies are then pullbacks of one differential object. This gives common phase provenance. It does not make the spaces X_i identical or make every operator acting on them the same.

An operator comparison needs another square: the line pullbacks must tensor the relevant state bundles, and the sector operators must intertwine the pullback maps. That extra operator square is exactly what the current q79 finite result supplies.

3 The exact q79 finite shared-line result

The selected finite branch uses a degree-three sheet system with monodromy in S_3 . Its abelianization has one nontrivial sign character. The corrected Foundation maps that sign into the order-two element of a universal flat $\mathbb{Z}_{64}$ differential line. In the Foundation’s notation,

$$h_{S_3}(\sigma) = 32 \epsilon(\sigma),$$

where $\epsilon(\sigma)$ distinguishes even and odd permutations. For both admissible odd roots, the resulting character agrees with the determinant sign. This produces a parallel isomorphism, not merely a fiberwise one: connections and every loop holonomy agree.

The same pulled-back line L_{sh} then tensors the three trace lanes,

$$\mathcal{H}_{\text{CLN}} = {}^{\dagger} \otimes (\mathcal{O} \oplus \mathcal{A}_0 \oplus \mathcal{A}),$$

whose ranks are 1, 2, and 3. Scalar phase transport commutes with:

1. the trace, trace-free, and full projectors;
2. the root-plane quarter-turn J_{DE} ;
3. the finite Haar or Reynolds projector; and
4. the normalized finite Hessian $H_{\text{fin}} = \kappa_{\text{fin}}(I - P_{\text{Haar}})$.

On the selected two-copy sheet/edge representation, the finite calculation gives

$$\text{rank } P_{\text{Haar}} = 2, \quad \text{spec}(H_{\text{fin}}/\kappa_{\text{fin}}) = \{0^{\times 2}, 1^{\times 4}\}, \quad H_{\text{TT}} = \kappa_{\text{fin}} I_2.$$

Tensoring by L_{sh} leaves these relations unchanged. No dimensionless fit is introduced to make the phase and Hessian data compatible.

3.1 Why this is a real advance

Earlier papers often said that several sectors “reuse a circle.” That could have meant merely that each sector had some isomorphic copy of $U(1)$. The finite q79 theorem instead identifies one differential source, its connection, its holonomy, and its commuting action on the relevant finite operators. It is a same-source result.

The theorem is exact at flat differential-character and finite-symbol tier. The physical HYM bundle has nonzero characteristic curvature, so its connection cannot literally equal this flat line connection. The next continuum result must be an intertwining or spectral-symbol functor that respects the different curvature data.

4 Lens, Nil, and the shared-circle language

Lens and Nil geometries remain useful, but not as literal nested containers. Both can arise from the Boothby–Wang or prequantization circle-bundle construction over different bases. Over a two-sphere, an integral circle bundle can give a lens space; over a two-torus, it can give a Heisenberg nilmanifold. Their connection curvatures and total-space topologies differ.

The reconciled picture is therefore:

$$\begin{aligned} &\text{flat q79 line: finite sign/phase transport,} \\ &\text{Lens and Nil bundles: curved circle-bundle realizations,} \\ &\text{physical HYM bundle: nonflat holomorphic gauge data.} \end{aligned}$$

These objects share a construction language through differential $U(1)$ -bundle geometry. They are not one manifold, one connection, or one extra coordinate. Literal $S^1 \times \text{Lens} \times \text{Nil}$ and literal circle-inside-Lens-inside-Nil nesting are not used as the physical q79 compactification.

5 Mass: what a circle can and cannot supply

5.1 The physical definition comes first

In relativistic quantum theory, a stable particle mass is read from the energy-momentum representation or, in a field description, from the location of the appropriate propagator pole. For unstable states one uses a complex pole or another explicitly declared scheme. A running Lagrangian parameter, a Hessian eigenvalue, and a physical pole mass are not automatically the same number.

Compact phase data can influence masses. Boundary conditions or holonomy can shift the spectrum of a Dirac, Laplace, or covariant derivative operator. Dimensional reduction can then turn an internal eigenvalue into a four-dimensional mass parameter. But the complete chain must be displayed:

$$\begin{aligned} \text{line and connection} &\longrightarrow \text{coupled operator} \longrightarrow \text{eigenvalue or self-energy} \\ &\longrightarrow \text{dispersion relation or pole.} \end{aligned}$$

The finite shared-line theorem closes only the first arrow and its compatibility with a finite Hessian. It does not identify κ_{fin} with an observed particle mass.

5.2 Relation to the Standard-Model carrier

The current MTT Standard-Model program has a finite 27×27 carrier and declared Yukawa, threshold, and profile/value tiers. Those results can constrain an internal mass matrix. To promote a row to a physical mass one still needs the normalized four-dimensional kinetic term, renormalization scheme and scale, self-energy or matching transport, and the pole or dispersion calculation.

The shared line is a promising common source for phases and selection rules. The statement that it numerically generates all masses or Yukawa magnitudes is not presently proved.

6 Inertia: response is not phase alone

Inertia is the response of a physical state to changes in motion. For a relativistic point-particle model, the action

$$S_{\text{pp}} = -m \int ds$$

has a low-velocity expansion whose coefficient m produces the familiar $\mathbf{F} = m\mathbf{a}$ relation. Deriving that relation from the assumed action is standard. It does not derive m from a circle.

An MTT inertia theorem would need to start from the selected field or effective action, identify a localized state, and compute its response to an external force or to a change in four-momentum. The result would then have to agree with the mass extracted from the pole or dispersion relation in the same normalization.

This gives a sharp equality test:

$$m_{\text{response}} \stackrel{?}{=} m_{\text{pole}} \stackrel{?}{=} m_{\text{gravity}}.$$

The equal signs are physical theorems, not consequences of using one phase line in all three calculations.

7 Gravity: a common line is not the Einstein equation

The current MTT gravity program conditionally recovers the four-dimensional Einstein/TEGR tensor structure from a selected higher dimensional branch after truncation, Lorentzian, normalization, and stress-response hypotheses are supplied. That result is the proper source for gravitational dynamics.

The shared line may participate in the compactification, spin structure, phase transport, or anomaly data entering that reduction. It can therefore help enforce common provenance across matter and gravity sectors. It does not by itself imply:

1. the Einstein–Hilbert or TEGR action;
2. universal coupling to the full stress tensor;
3. equality of inertial and gravitational mass;
4. the Newton normalization; or
5. horizon entropy and information recovery.

Each item requires the four-dimensional action or response functional. The shared-circle interpretation of gravity as “coherence bookkeeping” remains a useful conceptual proposal only if the selected action later realizes it.

8 Time: phase, order, and clocks are different objects

Four parameters have appeared near the circle discussion:

1. the compact phase angle θ ;
2. a stabilization parameter τ ;
3. physical Lorentzian time t ; and
4. a renormalization scale μ .

They are not interchangeable. The phase is periodic. Stabilization labels a mathematical relaxation. Physical time belongs to a Lorentzian initial-value problem. Renormalization compares descriptions at different scales.

8.1 How a circle may participate in a clock

A periodic degree of freedom can serve as part of a clock if a physical Hamiltonian, clock observable, calibration, and method of counting or unwrapping cycles are supplied. Even then, a periodic clock generally gives only local or cycle-relative time unless additional nonperiodic information is available. The circle phase is therefore capable of synchronizing sectors, but it is not itself a global noncompact time coordinate [5, 6].

8.2 Projection does not automatically create an arrow

A many-to-one projection loses upper distinctions, but the induced lower dynamics depends on how the upper evolution acts on projection fibers. The correct quotient theorem is:

an autonomous lower map exists exactly when upper evolution preserves projection fibers; it is invertible exactly when the fiber equivalence is preserved in both directions.

Thus a noninjective projection can coexist with reversible lower dynamics. The old claim that the absence of a global right inverse creates physical time is withdrawn.

An arrow of time requires a selected branch or state, Lorentzian evolution, and a monotone physical record, entropy, or clock relation. The two complex-conjugate finite phase lifts can remain mathematically admissible without both being well-behaved physical universes. Selecting the observed orientation remains a separate problem.

9 Families and phase selection

The q79 construction has a degree-three sheet system, and its finite carrier uses the $1 + 2 + 3$ rank flag. The shared line carries the abelianized sign phase of the sheet monodromy and participates in the finite SpinC/Fourier return. These are exact finite statements.

Three fermion families do not follow from the existence of an arbitrary circle. Their proposed identification uses the selected degree-three cover, monodromy, bundle representations, and the Standard-Model operator map. A circle can support many characters and windings; it does not uniquely choose three. The family claim must therefore be sourced by the full q79 cover and operator construction, with the shared line supplying common phase transport rather than the family count by itself.

10 Current status of the unification proposal

Claim	Status	Required evidence
One finite q79 shared differential line	Exact	Foundation finite shared-line theorem
Compatibility with finite projectors and Hessian	Exact	Commuting tensor action on selected symbol
Continuation to physical HYM geometry	Open	Connection-aware continuum intertwiner with curvature respected
Circle determines particle mass	Open	Coupled operator plus pole or dispersion theorem
Circle determines inertia	Open	Localized-state response calculation
Circle enforces equivalence principle	Open	Same-source inertial and gravitational response equality
Circle is physical time	Rejected	Compact phase and noncompact time are different objects
Circle participates in a clock	Conditional	Hamiltonian, observable, calibration, and cycle handling
Circle alone gives three families	Rejected	Degree-three cover and operator map are essential
Shared line organizes all four readouts	Research proposal	One connection-preserving source through all required operators

11 The shortest route forward

The promising idea is not that one circle should be renamed as four physical quantities. It is that one differential line may feed four independently defined operators. A decisive

same-source program would construct:

1. a continuum map from the flat q79 line to the physical SpinC/HYM/Dirac data, preserving the declared connection relations;
2. a normalized four-dimensional Dirac or two-point operator and a certified physical mass extraction;
3. an inertial response functional for the same localized state;
4. the four-dimensional gravitational response to that state's stress tensor; and
5. a physical clock observable whose phase readout is related to Lorentzian time on a declared domain.

Agreement among these readouts would be a nontrivial unification because the same source would survive several different maps. Disagreement would locate the exact arrow where the shared-circle interpretation fails.

12 Discussion

The finite shared-line theorem is stronger than an analogy and weaker than a theory of mass or time. It says that the q79 finite lanes and Hessian square really do share one phase-transport source. That is valuable: phases, determinant signs, and finite operators no longer need unrelated circle copies.

The physical ambitions of the first manuscript remain interesting. A common line could help correlate flavor phases, spinorial closure, compactification, and effective response. But physics is attached to operators and observables, not to suggestive names. Mass must be a mass in a dispersion law or propagator; inertia must be a response; gravity must follow from a varied action; and time must be read by a physical clock within a Lorentzian dynamics.

13 Conclusion

The shared circle survives the corpus reconciliation in a precise form. It is one Hermitian differential line carrying compact phase and holonomy, counted once, with an exact finite q79 intertwining action. It is not an extra dimension, literal compact time, or a direct derivation of mass, inertia, gravity, and families.

The unifying research program is now testable: transport the same line into the continuum operators that independently define each observable and check whether their normalized readouts agree. That is the correct way to turn shared bookkeeping into shared physics.

Computational Evidence and Reproducibility

The shared-circle paper uses the selected q=79, finite-matrix, HYM, anomaly, threshold, and electroweak rows as concrete instances of common phase or holonomy data. These packets do not identify the compact circle with physical time or derive inertia, mass, or gravity. The profile rows are context and the strict no-knob upgrade remains open.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json

Manifest SHA-256:

fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper’s local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/direct_k_higgs_row (*derived exact*).
Promoted direct $K_{\text{threshold}} \Omega_H \lambda$ row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A11/q79_exact_audit (*derived exact*).
Executable $q=79$ exact-branch audit.
- A11/q79_exact_theorem (*derived exact*).
CRT $q=79$ theorem on the selected exact branch.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27×27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_{EW} source row at the declared one-shared-primitive standard.

Corpus-state cross-checks.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Y_u, Y_d, Y_e and λ_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8×8 covariance.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

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